1 Introduction

1.1 Definition of masting

1. Observations of masting in relation to seed predation and flowering

1.2 Different hypotheses operating at different reproductive stages like filters

1. Pollination success (Pollination coupling)
2. Resources availability (Resource Matching)
3. Dispersal (Predation satiation)

1.3 Why traits can help link hypotheses to masting mechanisms

1. Mechanisms might not be mutually exclusive, but some might be stronger than the other for certain species
2. Traits thus can be important to understand masting through their relative importance

1.4 Trait-based perspective

1. Literature review on understanding masting from traits perspective
   (a) Wind pollination is related to higher CV
   (b) Animal dispersed seeds have higher CV than wind dispersed seeds
   (c) Larger seed is associate with higher CV
   (d) Recent papers also focus on resource perspective using leaf and shoot traits
2. Existing studies mainly trying to find relations between different traits and masting.
3. Not clear connection between traits and hypotheses and potential evolutionary drivers

1.5 Introduce framework linking hypothesis x traits x masting

1. Link traits to hypotheses help understand drivers of masting for different species
2. Especially when consider whether the evolutionary drivers are more similar for clades those are closely related.

1.6 Use data from one specific geographical region

1. Here we use North American tree data to test this framework since they have information on their life-history
   (a) Masting info available
   (b) Includes both masting and non-masting species
   (c) Not using CV data because don't have stand-level traits data

2 Discussion

2.1 Summary of main findings

1. Pollination supports the pollination coupling hypothesis
2. Seed weight supports the predation satiation hypothesis
3. These act at the start and the end of the reproductive cycle, provide evolutionary incentive, which I will explain below
4. Discuss mechanisms in the context of existing literature
   (a) Previous literature on seed mass and masting think from the resource depletion perspective (Bigger seeds use up resources so cannot reproduce every year)
   (b) However, from evolutionary perspective, seeds attractive to animal dispersers will benefit more from masting
5. Other traits show some pattern but statistically non-significant
6. Transition to alternative hypotheses

2.2 Masting is a population-level phenomenon

1. Other traits may matter to masting below the species level, like population level
2. Examples of species that mast only under certain environmental conditions (E.g., *Hymenaea coubaril*)
3. Traits need to be measured locally and match with masting data

2.3 Different species selected under different hypotheses

1. Introduce flowering masting vs. seed masting
2. Different species driven by different mechanisms
3. Need a more refined definition of masting and additional flowering-related traits
4. Transition to phylogeny

2.4 Evolutionary scale

1. Angiosperm vs. gymnosperm results
   (a) More masting gymnosperm sp
   (b) Longer leaf longevity in gymnosperm
   (c) More monoecious species in gymnosperm
   (d) Wind pollination is universal among gymnosperms
   (e) Gymnosperm and angiosperm have distinct trait space occupancy
   (f) Importance of looking for potential evolution support between gymnosperm and angiosperm
      i. Test if masting was gradually diluted in angiosperm as they transitioned to biotic pollination and faster life histories
2. Many traits have strong phylogenetic signal, it is hard to tease out traits from masting, which limits the inference
3. Data is strongly nested in phylogeny, and data is sparse.
4. Importance of testing within clades and collect more data (*Quercus.*)